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Research Statement

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My primary research focus revolves around data and algorithm efficiency problems that stem from addressing scalability and efficiency issues in the training and deployment of large-scale models. I employ tools from **non-convex optimization** and **coreset/subset selection literature** to address these bottlenecks effectively.

Industry reports highlight the rapidly growing computational demands of large-scale AI. McKinsey (2025) [NPS25] projects $6.7 trillion in global data-center capital expenditure by 2030, and Gartner (2026) [Gar26] links rising AI infrastructure spending to sustained demand for LLM training and enterprise deployment. Together, these trends motivate memory- and compute-efficient methods that preserve model quality, alongside a core requirement of designing verifiable and reliable AI models.

Key Research Theme and Tools: *"How can we make LLM training and adaptation more data-, memory-, and compute-efficient by designing principled subset-selection and weighting strategies, while retaining rigorous guarantees on performance and reliability?"*

• **Data efficiency:** selecting small, information-rich subsets that preserve downstream quality while reducing per-step compute and memory during both pre-training, mid-training and post-training phases.

• **Reliability and uncertainty:** developing verifiable procedures that remain robust under distribution shift and limited evaluation.

Pretraining and finetuning LLMs with billions of parameters is computationally intensive, motivating multiple works in parameter-efficient adaptation and training strategies.

These challenges have led to solutions via several paradigms including using smaller set of parameters [ALHS21], low-rank representations [HSW+21, ZZC+24], quantized weights [DLBZ22], or even gradient-free optimization for training language models [MGN+23]. However, a more nuanced (and largely complementary) lever is to improve the *quality* of the data seen by the model at each step. A core hypothesis is that selecting better training examples/mini-batches can yield substantial gains in sample efficiency and often improved generalization.

My research makes progress on this front from both a theoretical and an applied lens. I formulate *structured* data-selection and training objectives with appropriate constraints/parameter-sharing to make the problems statistically meaningful and computationally tractable, and I design practical algorithms with rigorous guarantees. The resulting formulations often involve **non-convex structure** (e.g., **low-rank structure, sparsity, latent mixture structure**), and thus require a careful blend of tools from **non-convex optimization** and **statistics**. **Some of these techniques were prototyped in internal Google models during my 2024 DeepMind internship on Uncertainty in LLMs, improving data efficiency in training.**

In the subsequent sections, I describe the main themes of my research in more detail and outline several related directions that I am excited to pursue going forward.

Data Efficient Training of Large Language Models

Typical training of large language models (LLMs), during either pretraining or post-training, requires substantial GPU memory to store model weights, gradients, activations, and optimizer states. For instance, full-parameter training of Llama-2 7B [TMS+23] requires *approximately 112 GB of memory* just to store the model states, without accounting for activations and temporary buffers.

Since activation memory scales with batch size and sequence length, training with large batches often necessitates techniques such as model-state sharding, activation checkpointing, or CPU offloading. These memory constraints limit the use of larger batches and greater data parallelism, which can improve hardware

utilization and reduce gradient variance. This has motivated growing interest in developing more effective minibatch selection strategies for language model training [NYA+25, CAG+26, ACK+26].

Prior work [KDR+21, KSRI21, MBL20] studies gradient-based data selection, mostly for vision models. As models scale, these methods become expensive because gradient dimensionality grows with the number of parameters, making them particularly impractical for LLMs.

Many methods therefore pose minibatch selection as an *optimization problem*, choosing samples whose gradients best match the full-batch objective (or improve validation performance). More broadly, **scalable and efficient training remains a major infrastructure and compute challenge**.The main questions here are: *"Can we design effective data selection algorithms to train LLMs with smaller informative mini-batches and get similar benefits of training with full batch sizes, with no extra training overhead time? Can strategies e.g. curriculum learning, reduction in gradient conflicts between domains be implemented in conjuction with the selection algorithm?"*. Answers to these questions can lead to significant reduction of the *memory compute* and *better downstream generalization*.

I study the problem in two distinct learning environments. *(1)* **Sequence Learning:** We study how the ordering of training samples influences optimization dynamics, convergence, and downstream performance—a phenomenon widely explored in the curriculum learning literature. *(2)* **Static Learning:** We focus on selecting informative instance-level mini-batches at each training step, without explicitly modeling long-horizon ordering effects.

A key challenge in designing offline curriculum-style training strategies is that *(1a)* partitioning the entire training corpus into domains based on difficulty is costly, while *(1b)* assessing sample difficulty often requires domain expertise. We address *(1a)* by developing *dynamic (on-the-fly)* curricula that automatically identify easy vs. hard minibatches and schedule them during training, and we tackle *(1b)* by using model-uncertainty signals as a scalable proxy for example hardness.

Designing subsets for Curriculum Learning: In the context of *(1a)*, [KEP+23] (`MILO`) reports an intriguing observation in an *offline* setting: *representativeness*-based submodular objectives tend to select *easier* training examples, whereas *diversity*-based objectives tend to surface *harder* examples. In [CAS+26], we further validate this effect and treat these objective "signatures" as useful proxies for curriculum design. To move beyond the restrictive offline assumptions made in `MILO`, we cast curriculum learning as an *online* multi-armed bandit problem: each submodular objective (from the *representativeness/diversity* family) is an arm, and a validation-guided reward adaptively chooses which objective to deploy throughout training. This adaptive strategy yields significant improvements over `MILO`'s fixed mapping ("representative $\Rightarrow$ easy", "diverse $\Rightarrow$ hard") and over other computationally expensive bilevel selection objectives (e.g. [KDR+21, KSRI21, WWS+24], across both vision and language settings.

Designing Efficient Utility Functions: A second goal, in the context of *(1b)*, is to design utility functions that select informative samples from each training batch. Since [CAS+26] defines a validation-guided reward over a subset of the minibatch, we redesign this utility : shifting from a discrete sample-selection problem to a continuous importance-weighting (importance sampling) formulation. This shift is theoretically advantageous: in the continuous regime, each training example in the batch is assigned a non-negative importance weight, and the resulting utility is ***weakly*-submodular**. This enables the use of efficient algorithms such as *orthogonal matching pursuit*, adding little to no overhead to training. Finally, when constructing these utility objectives, we consider different constraint families that respect the order and mechanism by which samples are selected.

Our work [ACK+26], for instance, considers a **partition matroid constraint**, where each domain represented in the training batch is subject to a domain-specific capacity (budget). These budgets can be tuned by the practitioner and can thus be viewed as a coarse mechanism for domain-mixture design. In contrast, traditional data-mixture methods (e.g., [XPD+23, LZM+25]) typically train many small proxy language models (often on the order of 100–200) and fit a regression model to estimate per-domain mixture weights. Relatedly, in [CSKR26] we study mixture learning for supervised fine-tuning over heterogeneous tasks: `TaskMixPGM` models tasks as nodes in a Markov random field, combining unary *utility* terms with pairwise

interaction potentials (via behavioral divergences) to learn continuous mixtures and derive weakly-submodular discrete selection under budget constraints.
As a practical alternative, we show that setting capacity budgets directly from the training batch's observed domain composition can outperform these baselines, while assigning greater importance to under-represented domains than **expensive data-mixture approaches**. We aim to study this more rigorously under a broader theoretical framework. In particular, our main result in [ACK+26] shows that domain-aware selection reduces inter-domain gradient conflicts within a minibatch, leading to a more diverse selected set and improved training dynamics.
More broadly, *gradient conflicts across domains* can arise at many points in an LLM's training lifecycle. In **continual learning**, this interference is especially problematic, since learning new tasks can overwrite prior capabilities and lead to **catastrophic forgetting**. To address this, in our joint work [NKH+26], we propose `OSFT`: a parameter-efficient continual-learning method that uses adaptive SVD to identify and preserve high-rank subspaces encoding prior knowledge, while constraining new-task updates to be strictly orthogonal to them. This reduces interference without adding parameters or storing task-specific gradients. Across standard continual-learning benchmarks (`T5-Large`, `LLaMA-2 7B`, `Mistral-7B`), `OSFT` achieves a strong learnability–retention trade-off, improving average accuracy by up to 7% over baselines such as `O-LoRA` and reducing forgetting to near-negligible levels, while preserving general capabilities, instruction-following, and safety.

Fairness in Domain Representations: Large language models are commonly trained on corpora that mix many domains, but the resulting domain composition is often highly imbalanced. For example, *MetaMathQA* [YJS+24] contains data from multiple domains with a strong emphasis on mathematics. Under such imbalance, standard minibatch construction and subset-selection heuristics can systematically under-sample minority domains, leading to biased training signals and poorer generalization on underrepresented domains. This connects naturally to our `PartitionSEL` line of work, where we enforce explicit per-group selection budgets via partition constraints. Complementarily, in our work [CAG+26], we address domain imbalance by casting per-batch subset selection as a *partial optimal transport* problem: we pick a fixed-size set of prototypes with *uniform* weights and transport their mass to match the target distribution subject to capacity constraints. We instantiate the OT cost via a gradient-similarity kernel, using pairwise dot products between training samples' last-layer gradients. This yields a tractable submodular objective with a greedy $(1 - 1/e)$ approximation guarantee, and empirically improves minority-domain coverage without sacrificing performance on majority domains.

Role of Spectral Optimizers in Gradient Matching based Coreset Selection Adam [KB14] has been the all time state of the art optimizer for a majority of deep learning advances till date, with strong empirical benefits across multiple architectures and different modalities (e.g. text, speech, vision). However, with Kimi K2 release, spectral optimizers such as *Muon [JJB+24, BN24] has shown significant better gains than Adam/AdamW counterparts*. Existing coreset methods do not account for the anisotropic, matrix-valued geometry induced by spectral optimizers such as Muon [JJB+24]. In our work [CSG+26], we show that Muon's momentum-matrix orthogonalization can better exploit this anisotropy than Adam-style updates: without it, a few dominant (and often noisy) gradient directions can disproportionately steer the update and degrade the quality of the per-step selected samples. Orthogonalization instead normalizes the momentum spectrum by keeping all eigenvalues at one, effectively treating gradient directions more uniformly.
All contributions referred to above appeared in NeurIPS, AISTATS, ICLR and ICML.

Vision for future research.

- **Optimizer-aware gradient matching.** In ongoing work, I argue that gradient-based coreset methods make two separable design choices: (i) the *geometry* used to measure mismatch and (ii) the *target* the subset is asked to approximate. This separation helps explain why methods with similar conditioning can still behave differently, and it motivates a central question for adaptive and spectral optimizers: *When does matching the full-batch gradient also match the update the optimizer would take?* For a fixed linear

preconditioner, update error reduces to the norm of the *preconditioned* gradient error; with momentum, adaptive preconditioning, or Muon-style orthogonalization, the impact of a gradient error becomes state- and batch-dependent. I aim to characterize which error components these transformations *amplify* or *suppress*, and use that analysis to design selection objectives that approximate the *parameter update* without computing it on the full batch. By varying geometry and target independently, I want to isolate when gains come from being optimizer-aware versus from targeting an update that better aligns with validation performance.

- **Data selection for efficient and verifiable unlearning.** Inspired by the link between subset selection and machine unlearning highlighted in Kamath's ICML 2026 keynote [SKW26], I want to design selection rules that remain faithful under future deletions. A subset can match the original-batch update yet misrepresent how that update would change if a user's records (or an entire source) were removed.
  I plan to develop objectives that control update error over a family of group deletions and to identify when a lightweight *post-hoc correction* can approximate retraining on the retained data. Using retraining as the reference, I will evaluate fidelity and cost, with emphasis on underrepresented domains, and clarify what per-example statistics can be retained under privacy constraints.

Robust Generalization and Verifiable Reasoning

In the preceding work, I use signals available during training, such as validation performance and gradient alignment, to decide which examples should contribute to an update. This raises a related question about supervision itself. Once an example has been selected, how much weight should the model give to each source of supervision? At inference time, a similar decision arises when a model generates a multi-step answer. Which intermediate steps warrant a check, and when should the model revise its reasoning? In both settings, I use information available during the computation to decide where an intervention is worth making.

Adapting supervision under distribution shift. In our work on knowledge distillation [SCR26], standard distillation is already a strong baseline and can outperform specialized domain-generalization methods. Its use of a fixed distillation weight, however, treats the teacher's predictions as equally useful for every example. We use early-layer representations of the student to train a small network that adjusts the balance between teacher supervision and ground-truth labels for each instance. On vision and text benchmarks, this improves out-of-distribution performance over standard distillation and empirical risk minimization, with no additional inference-time computation. The work extends the idea of instance-level selection from deciding which examples to train on to deciding how to learn from each example.

Verifying reasoning as it unfolds. In InterWhen [BCE+26], we study how to steer a reasoning model toward outputs that satisfy task constraints expressed in natural language. A final answer can appear correct even when the reasoning violates a constraint along the way. Our framework checks intermediate reasoning against the task's requirements and uses verifier feedback to guide the model when a violation is detected. Checking an incomplete trajectory creates an opportunity to correct an error before subsequent steps build on it. This gives us a way to allocate inference compute to verification and repair, and to compare that allocation with spending the same budget on additional candidate solutions.
All contributions referred to above appeared in NeurIPS, TMLR, and ICML.

Vision for future research.

- **When (and how) guidance transfers across tasks.** I want to understand when external guidance helps a model generalize to unfamiliar tasks. In our distillation work, the student's early representations help determine how much weight to give the teacher on each example. I want to extend this idea to adaptation with few labeled examples. Task affinities can suggest where to look for useful supervision, while the student's representations may reveal which of those sources can address its current errors. By measuring how target performance changes when we vary supervision from related tasks, I aim to understand when sharing improves generalization and when it reinforces existing errors.

- **Which reasoning steps are worth verifying.** The same question arises during reasoning, where a limited verification budget forces a choice about which steps to check. Building on InterWhen, I want to identify decisions where feedback can change the outcome. Some errors resolve during unaided continuation, while others become harder to repair as later steps build on them. Starting from the same partial trajectory, I would compare unaided continuation with verifier-guided revision to estimate where an early check is most valuable. The goal is to allocate verification to steps where it is most likely to improve final-answer accuracy and adherence to task constraints under a fixed inference budget.

Use of AI for this Research Statement:

I personally felt that spending more than two days on polishing my research statement may not be a good use of my time, given AI would have been able to write this well for me in a jiffy and also given the fact that at this point in time it has already solved millennium prize problems. But in hindsight, writing this manually was a cathartic experience, given it opened up several gaps in how I desired to communicate the problems I have been working on. Apart from writing the content in my own words, I used Overleaf Premium for LaTeX editing and formatting, and for occasional sentence-level rephrasing.

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